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Contagious ideas and cognitive artefacts

the SWOT Analysis evolution in business

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Title: Contagious ideas and cognitive artefacts: The SWOT analysis evolution in business

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Summary: This historical review uncovers the institutionalisation and diffusion of the SWOT analysis by assessing academic literature, seminar materials, proprietary research reports and interviews with experts from the virus theory perspective. We suggest that reviews of the SWOT analysis using the management fashion theory perspective are inadequate in explaining the diffusion and rejection of ideas born in practice. The virus theory perspective starts at an organizational level and reveals that predominantly practitioners were instrumental in spreading the ideas like participatory planning and distinguishing between short term and long range planning in order to resolve the planning paradox in provisional planning. Due to mutations in practice by consulting firms, the 2x2 matrix of SWOT became a cognitive artefact on its own. Theoretical roots of the original SWOT analysis stem from psychology and behavioural sciences. It is questionable if current strategy textbooks reflect these theoretical backgrounds.

Word count: 4966

Introduction

The use of SWOT analysis continues to permeate the academic peer-reviewed literature (Meyer, Cohen & Nair 2016; Shantz, Podolsky & Frawley 2016; Bell & Rochford 2014 and Helms & Nixon 2010) and is still featured in prominent strategy textbooks (Rothärmel 2017; Johnson 2017). SWOT analysis, one of the oldest strategy tools, is ubiquitous in practice (Knott 2008; Gunn & Williams 2007; Glaister & Falshaw 1999) and has become part of everyday language. SWOT analysis is even a subject term of both Business Source Premier (EBSCO host) and Business Premium Collection (ProQuest) thesauri.

To understand how things – ideas or practices – get from here to there scholars have developed several approaches to explain the diffusion of management ideas, however there is insufficient research on the role academics and practitioners play in the evolution and diffusion of cognitive artefacts embodying contagious management ideas (Jarzabkowski & Kaplan 2014; Katz 1999). Understanding where the SWOT analysis comes from and why it was developed may contribute to the research agenda of strategy as practice (Jarzabkowski, Spee & Smets 2013; Jarzabkowski & Spee 2009) and organized entrepreneurship in particular by uncovering forgotten ideas. If we are to look at the institutionalization of strategy tools at the field level, the historical methods such as those used to analyse management consulting and other management practices is called for. A field-level mapping of the evolution of strategy tools over time would give insight into how strategic management is conceived as an activity and a profession (Jarzabkowski & Kaplan 2014).

Research question and contributions

This article reports on the historical context, infection and diffusion of the contagious idea called SWOT analysis and assesses why, how and by whom the SOFT approach, which was developed to solve the so-called planning paradox, originated and evolved into the SWOT analysis. The paradox is this*: In the absence of defined plans, how can one appraise operational effectiveness? But, without knowing how effective current operations are, how can one create a strategy regarding what to keep and what to change*? (Stewart, Benepe & Dosher 1965). The SWOT analysis gradually mutated once it got into practice. The theoretical roots will be discussed and in doing so, we are able to understand how a cognitive artefact evolves and is proliferated within organizations in which both academics and practitioners played an active role.

The research question is: How did SWOT analysis evolve into a dominant cognitive artefact in management**?**

By answering this question, we aim to develop insights and make the following contributions:

- Further develop the virus theory by the analyzing the origin, diffusion and institutionalization of a contagious management idea from an organizational perspective.
- Contrast the findings from the virus theory perspective with the management fashion theory. What do we learn from virus theory perspective about the institutionalization of contagious management ideas and how did various actors help or hinder the evolution of SWOT analysis?
- Develop an understanding of SWOT analysis originally was designed for. To what degree are the current strategy text-books in line with the assumptions of the developers?

Theoretical framework

To assess the evolution of the SWOT analysis we use the 'virus-inspired theory' or 'virus theory' (Røvik 2011). The virus theory has six viral features which correspond with ten organizational idea handling processes in light of insights about what viruses are, how they spread and how they function in active hosts. The viral features are infectiousness, immunity, replication, incubation, mutation and dormancy. Infectiousness corresponds with the formal decision to adopt a management idea. A prerequisite is an active host. Next are the immunity mechanisms (non-adoption, isolation, expiry and rejection) which can inhibit the adoption of the management idea. Replication occurs when the entrenchment mechanisms kick in and the management idea becomes anchored in organizational structure, routines and daily activities. After a while incubation sets in maturation of the management idea leads to a slow-phased transformation into practice. Mutation occurs when attempts are made to translate the ideas for implementation in practice. The dormancy of an management idea can occur because of inactivation of due to an upsurge, reactivation of dormant management ideas are turned on again.

The ten identified idea-handling processes are related to one another in numerous and complex ways. For obvious reasons, it is difficult to work out a theoretical representation that accounts for all the possible ways they can be connected. Even so, the relations can be classified according to three general types of relatedness which can exist between ideas: succession, tangling and competition (Røvik 2011).

Virus features	Organizational idea-handling processes
Infectiousness	*Adoption*
Immunity	*Non-adoption*
	Isolation
	Expiry
	Rejection
Replication	*Entrenchment*
Incubation	*Maturation*
Mutation	*Translation*
Dormancy	*Inactivation*
	Reactivation

Table 1 Viral features and corresponding idea-handling processes

In his article from fashion to virus (Røvik 2011), he already anticipates negative associations of the virus metaphor to diseases and even death. The outlined theory is neither founded on this, nor on other normative premises. It helps to identify relevant organizational idea handling processes.

Methods and data

The search process

In the scoping search we scanned both Business Premium Collection (ProQuest) and Business Source Premier in order to retrieve the scholarly articles, browsed the internet looking for websites and information on SWOT analysis and sent out an email requests to the Stanford Research Institute Alumni web form, based on a letter published in the SRI alumni newsletter December 2005 (Humphrey 2005). If articles were not available in both collections, we requested hardcopies via the interuniversity library lending service.

Citation analysis

We built and MS Access 2013 database for our own citation analysis, based on the most elaborate review article on SWOT analysis (Helms & Nixon 2010). Our preliminary findings pointed us to two books as sources of the SWOT analysis. We bought both Business Policy – Text and Cases (Learned, Christensen, Andrews and Guth 1965) and Corporate Strategy: An Analytical Approach to Business Policy for Growth and Expansion (Ansoff 1965) on Amazon.com for review.

Archival research and data collection

Our historical archive is a built from a collection of sources. Our first historical find was the Guide to SWOT analysis (Mottershaw 1974) which we retrieved from the National Library in Dublin, Ireland. The authors got in contact with dr. Don Nielson via email and were fortunate to receive his personal email archive for his book (Nielson 2004) via dropbox. He put us into contact with Bill Guns who sent us the crucial LRPS reports no. 250 via email. In August 2016 we met with professor William D. Guth at the Academy of Management and talked about the origin of SWOT Analysis in relation to the Harvard Business School. In some follow up interviews, we discussed some early strategy literature, the academic climate in the ‘60’s of the last century and looked at archival information in the Baker Library Historical Collections in Boston, Massachusetts on the Business Policy meeting in held at Harvard University in 1963, based on a reference (Ghemawat 2002) by professor Pankaj Ghemawat. We requested hard-copies from several folders. Next, we reached out to professor Ghemawat for clarification of this claim. He referred to a conversation with Kenneth Andrews, but his detailed notes were not available at the time.

After attending the Academy of Management in 2016, the authors visited Bill Guns at the former SRI site in Menlo Park. We talked about our research and he helped us with some key documents for our historical archive and shared his knowledge on the history of SRI in general and the TAPP group in particular. Around February 2017, we searched Facebook in order to trace former consultants of Urwick Orr & Partners in London. We had email contact Doreen Mortimer and Derek Hindley, who provided us some clues on Urwick Orr & Partners publications. In March 2017 we perused the online files from the Catalogues of Copyrights and inquired with the Library of Congress if hardcopies of LRPS reports we available. These efforts were to no avail.

The personal TAPP group archive from the authors with proprietary LRPS reports, seminar materials, surveys and correspondence has been compiled and structured in another MS Access 2013 database.

The authors have written permission to use copyrighted SRI materials from the copyright owner.

Dataset

A separate dataset will be published figshare, a repository to make research data where users can make all of their research outputs available in a citable, shareable and discoverable manner
For more details, please consult the appendix

Unraveling the history of SWOT analysis

Before analyzing the origin, diffusion and institutionalization from the virus theory perspective, we first want to discuss the current understanding of the SWOT analysis from in the academic literature. We want to address two major misunderstandings: the critique on the SWOT analysis by academics and the origin of the SWOT analysis.

First, scholars have critiqued the SWOT analysis for being an atheoretic classificatory system (Grant 2008), having a tendency to generate long (and often meaningless) lists of descriptive items (Glaister & Falshaw 1999) and ineffective as means of analysis of as part of a corporate strategy review. Some even claim that it is time for a product recall (Hill & Westbrook 1997). An extensive review of popular textbooks suggests deficient expositions of SWOT analysis abound (Valentin 2001).

Second, some prominent authors state that the SWOT analysis originates from Harvard Business School (Ghemawat 2002; Mintzberg 1994), while other sources point to Stanford Research Institute (Nielson 2004; Royce 1985). Our own citation analysis led us to the books Business Policy – Text and Cases (Learned, Christensen, Andrews and Guth 1965) and Corporate Strategy: An Analytical Approach to Business Policy for Growth and Expansion (Ansoff 1965). Both books are clearly not the sources. The most elaborate review article on the SWOT analysis (Helms & Nixon 2010) states that the origin is unknown. Internet sources on the history of SWOT analysis (Friesner 2017; Morrison 2017 and Humphrey 2005) agree that myths and citation errors are continually being repeated due to citation practice. The genealogy of SWOT is still unclear (Madsen 2016) and the quest for the primary source is not yet concluded (Bekdal 2014).

The laboratory for corporate strategy tools: the TAPP group at SRI (1962-1972)

The SWOT analysis started out as SOFT approach in order to solve the so-called planning paradox. This was published in the LRPS report no. 250 Formal Planning - The Staff Planner's Role at Start Up (Stewart, Benepe & Dosher 1965), describing in detail how formal planning can best begin with provisional planning.

The Long Range Planning Report Service (LRPRS) started in 1958 as a research program at the Stanford Research Institute (SRI). The word “Report” got dropped by Robert D. Bruce in 1960 to indicate the provision of something more than the printed reports for clients. The LRPS Staff were pioneers in the development of corporate strategy tools. The new service attracted attention because of its innovative character and quality of output. Between 1960 and 1962, some Fortune 500 clients of LRPS came back to SRI and said: *Now you are doing a good job of telling us what to plan for. But the truth is, we don’t know how to plan. Can you help us with that?* After much internal discussion and some talent searching, it was decided that it would be appropriate for LRPS to add a research activity to study how planning is done, analyse successes and failures, and to develop a leading-edge methodology for planning (Nielson 2004; Royce 1985).

Most of the SRI work was carried out for individual client companies and as a results were proprietary information. The generic innovations with which SRI helped systematize long-range corporate planning were themselves only slightly more public. The initial fee for the LRPS reports was something like $2,500 per year (Nielson 2004).

At SRI in 1961, Robert W. Smith, assistant director of Economic Research hands a copy of the paper An Organismal Theory of Business Planning (Stewart 1961) to William B. Royce, then assistant director of the Management Sciences Division.

The deficiency with planning lore today is not inaccuracy; it is lack of a standard frame of reference for practitioners. Take for instance the discipline of Accounting. The philosophy of double-entry, LIFO versus FIFO, double-declining versus straight line amortization, etc. have a common reference in the standard concept of a "set of books". The practitioner of planning does not have such a frame of reference. At least it does not exist in the general consciousness of those associated with the field – teachers, students, staff practitioners and managers. And yet, it seems to me, it does exist in the real world and needs only to be recognized to be useful.

Table 2 Excerpt from An Organismal Theory of Business Planning (Stewart 1961)

Smith suggested that Stewart might be the bridge person who could bring the management scientists and the industrial economists together in a coordinated approach for corporate strategy and planning. He worked at the time as senior member of the Corporate Development Planning staff at Lockheed in Burbank with director James Lipp, specialized in master planning (Stewart & Lipp 1962) and had a good reputation in planning theory in the industry (Steiner & Miner 1977; Steiner 1963). H. Igor Ansoff developed the concept of diversification when he joined Lockheed. Smith managed to lure Stewart away to SRI (Royce 1999) and in 1962, Stewart joined SRI and became the leader of a new LRPS program that was to spearhead most research at the Institute for the next decade called the *Theory and Practice of Planning* (TAPP).

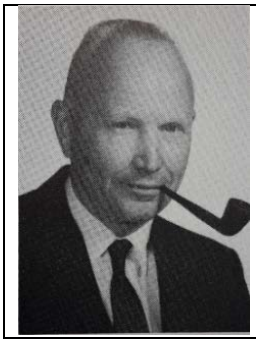
	Robert F. Stewart had industrial engineering (MSc. at UCLA) and psychology (AB) education, was a voracious reader, amateur musician and history buff. A renaissance man. He worked through college partly as an entertainer, partly in industry jobs. Practiced as an industrial engineer before joining the planning staff at Lockheed as a developmental planner. He wrote and supervised a dozen research reports on planning related topics, led some 50 seminars in business planning, consulted with numerous companies on design and implementation of planning systems and was a speaker at major planning conferences. He died in 1972.

Table 3 Profile of Robert F. Stewart, leader of the TAPP group 1962 - 1971

Evidence-based management was a secret to the success factor of the TAPP group

The TAPP group was a mixture of industrial engineers, business majors, economists, and psychologists (most with two or more degrees), all with industrial and some government and/or teaching experience. By 1964, the following key people joined. Industrial psychologist dr. Otis Benepe, who came from Lockheed, industrial engineer, Albert S. Humphrey came from Boeing, research-writer Marian O. Dosher from Westinghouse and industrial engineer Birger Lie, Norwegian Productivity Institute who scoured the literature as the TAPP group librarian. In 1966, industrial engineer Manuel Sotomayor from General Petroleum (today ExxonMobil) joined to help with computer simulations. Always looking from the outside in, they practiced evidence-based management (Rousseau 2012) *avant la lettre*. The TAPP group, headed by Robert F. Stewart, was never more than four to six but drew on other researchers throughout SRI. (then about 300) and some outsiders for the research reports.

Professor H. Igor Ansoff was one of those, he never worked at SRI, but was a frequent speaker on the SRI Executive Business Meetings and an outside academic consultant to SRI. He also authored several LRPS reports between 1967 – 1978.

The SOFT approach for provisional planning to solve the planning paradox

In his early work on creativity in business planning, Stewart quotes Colonel Lyndall Urwick, he says: "*The most telling measure of a managers ability is the degree to which he can elicit teamwork from his faithful administrators from his bastard geniuses*" and assure that they enter the game when their speciality is required (Stewart 1960). He draws inspiration from the principles of management (Koontz & O'Donnell 1959) and psychometry (Guilford 1956)

There are two important changes in the process of reaching business decisions. The business leader traditionally has been responsible for, and able to do, most of the thinking ahead for his firm. But the increasing size and complexity of today's businesses, plus the rapid changes in the environment, are forcing the chief decision-maker to request more and more help from other in assembling and analysing pertinent information. Planning is becoming a "participative process". The second kind of change is the firm's ability to select and shape the environment in which they will operate. Firms are not waiting anymore for opportunity to knock on the door. Motivation, options in the environment, and capability are the three domains surrounding decisions that define the firms values and prepare the circumstances that will assure motivation toward certain results in the future. The values that are the source of the organization's motivation are derived from stakeholders, a term coined by Marion Dosher (Freeman & Reed 1983; Stewart, Allen & Cavender 1963). A system is needed to subdivide the work into assignable tasks and to assemble the pieces of information that are obtained into a cohesive, valid chain of reasoning (Stewart 1965).

The virus theory perspective on SWOT analysis

To assess the origin, evolution and dispersion of the SWOT analysis, we use the six salient virus features with ten corresponding organizational idea-handling processes. Infectiousness, immunity, replication, incubation, mutation and dormancy. We describe these organizational idea-handling processes to understand the institutional dynamics around SWOT analysis.

The infectiousness of the SOFT approach

The first salient virus feature is the degree of infectiousness, which corresponds with the formal decision to adopt a management idea. Clients of SRI who attended the seminars and consultants acted as active hosts and intermediaries to spread the ideas, tools and techniques for provisional planning (Stewart, Benepe & Dosher 1965; Lie 1965). External experts (academic or CEO's) talked about different topics on planning. The participants were all subscribers to the LRPS reports and received all the training materials.

A Framework For Business Planning (Stewart 1963) was the first report for LRPS that Stewart wrote became a landmark publication (Mintzberg 1994; Steiner 1979; Steiner & Miner 1977; Ansoff & Stewart 1967; Learned, Christiansen, Andrews & Guth 1965). It laid out the pattern for the series of reports on the planning framework that followed the next 8 years. The 'SRI System of Plans' was widely copied and adopted by companies and consultants (Royce 1985). These reports where the architecture for a planning framework and described what to plan for.

For the Business Executive Seminars, which were boot camps for clients of TAPP who came to train with their companies how to plan, were organized between 1965 and 1971. Robert W. Smith, Robert F. Stewart, Albert S. Humphrey and Robert D. Bruce designed seminar materials and wrote an elaborate fictitious case study, called The Quinta Corporation, a case study (Stewart, Humphrey & Smith 1965) to learn about the theory and practice of planning.

Immunity for the SOFT approach

The second salient virus feature are immune reactions, which can be triggered as defence mechanisms, after the formal decision has been made to adopt a management idea (the infection). Subscribers to the Long Range Planning Service knew from the LRPS reports *what to plan for* and participants of Business Executive Seminars learned *how to plan*. However there has been very little empirical study on how organized, formal, planning is practiced. A LRPS report (Ringbakk & Dawson 1968) was commissioned to study the experiences of 40 large US companies with organized, or formal, planning and with a designated planning specialist with as much candour and realism as possible. The results were published in Long Range Planning (Ringbakk 1969). These companies were all SRI clients and most of them had attended one or more Business Executive Seminars. The findings in this study suggest that organized corporate planning is in the early stages of evolution. Management ideas like the SOFT approach are still done in isolation in the C-suite. Company strengths and weaknesses are evaluated in the light of external opportunities and threats. The extent to which this was done varies widely. This study made no attempt to determine quality of the different facets of planning; however, it may be noted that few companies have been able to integrate systemic surveillance of the environment with a management information system (Ringbakk & Dawson 1968). This is an indication outer defence mechanism of organizations known as *non-adoption*.

We know from letters in the Kenneth Andrews Papers, Harvard Business School Archives at the Baker Library that Albert S. Humphrey contacts professor Kenneth R. Andrews in 1967 to inform him about the SRI material on formal business planning (Humphrey 1967a).

He suggests that the dean of the Harvard Business School should attend one of the upcoming seminars so that Harvard Business School could use the material in their MBA course, including teaching materials. He goes on to say that to his knowledge no business school is practicing formal business planning or has a course for the same. As an alumnus, he submits Harvard be the first to do so. Andrews responds (Andrews 1967) that he finds the offer attractive, but asks for a waiver of the tuition component of the seminar, which was $ 1.850. He also indicates that the Business Policy course at the Harvard Business School embraced the topics outlined in the SRI seminar brochure and suggests to read the book Business Policy – Text and Cases (Learned, Andrews, Christensen & Guth 1965). Humphrey replies that he will try to engage George Steiner from UCLA to help try to convince the directors at SRI to release their materials for his research purposes (Humphrey 1967b). This is an example of *rejection* because of a monetary issue (tuition fee for a seminar in order to obtain all the seminar information to teach an MBA course). Teaching materials circulated among academics and is also in line with observations by professor William D. Guth (Guth 2017) who said that SWOT analysis was *all of a sudden out there* and has been part of the intellectual atmosphere ever since.

Replication

The third salient feature of a virus is the ability to reproduce which corresponds with the entrenchment of an management idea in practice. The entrenchment and effects of management concepts may be facilitated or hampered by the amount of resources devoted to educating and training organizational members in various aspects of the concepts (Røvik 2011). In 1970, all 600 graduates from the 300 organizations that had attended the Business Executive Seminars received a survey (Stewart 1970) in order to write a progress report on their experiences with the SRI approach for a planning system. At this time, the results from this survey are not available to the authors. But, we can establish that the implementation of the SRI planning philosophy was being monitored.

The abridged overview (see table 4) of the ISIC (UN 2017) sectors from which the companies came to learn the SRI philosophy of planning gives an impression of the corporations who came in contact. Many of them are large, multinational corporations spanning many sectors. This could also be an indication of the *entrenchment* and longevity of the management idea.

ISIC	Description of the sector	SRI Clients who attended the Business Executive Meetings
H51	Air transport	Air Canada, Lockheed Aircraft Corporation, United Airlines
D35	Electricity, gas, steam and air conditioning supply	Allgemein Elektricitäts-Gesellschaft, Illinois Gas Company Tennessee Gas Transmission Company
M06	Extraction of crude petroleum and natural gas	Esso Chemical S.A., Esso Standard Italiana, Preussag AG Standard Oil Company
K64	Financial service activities, except insurance and pension funding	A. Sarasin & Cie (Private Bank) AG, Ab Nordiska Förenings-banken, Algemene Bank Nederland N.V., Bank of Ireland Group, Chase Manhattan, Chesham Amalgamations and Investments Limited, Continental Illinois National Bank and Trust Company of Chicago, Den Danske Landmandsbank Aktieselskab, Der Norske Creditbank, Enskilda Bank, First National Bank, Salink Bank, Stockholms Enskilda Bank, The Canada Trust Company
I56	Food and beverage service activities	California Packing Corporation, Canteen Corporation, The Coca-Cola Company
K65	Insurance, reinsurance and pension funding, except compulsory social security	Continental National American Group, Swiss Reassurance Company
C21	Manufacturers of basic pharmaceutical products and pharmaceutical preparations	American Cyanamid Company, Warner-Lambert Pharmaceutical Company, F. Hoffmann-LaRoche & Co.
C28	Manufacturers of machinery and equipment	Colt Industries Incorporated, AB Karlstads Mekaniska Werkstaf, Bearings Operations, Case Company, Deere & Company
C17	Manufacture of paper and paper products	Appleton Coated Paper Company, Champion Papers, Domtar Limited, Temple-Eastex Inc. (Time, Inc. subsidiary), International Paper Company, The Bowater Paper Corporation Limited

Table 4 Abridged client list of SRI clients per sector who attended Business Executive Seminars (1965-1971)

Incubation

The fourth salient virus feature is incubation. Over time incubation sets in maturation of the management idea leads to a slow-phased transformation into practice. The SRI philosophy of planning has taken 8 years to complete and gradually came into practice. In American Finance: three views of strategy (Unterman 1974), a brief evaluation of the Stanford method describes the provisional planning phase based the experiences of top management of Chase Manhattan (at present part of JP Morgan Chase) from the TAPP group. In the same year, Getting Management Commitment to Planning – A new approach (Humphrey 1974) redresses the SRI approach and explains the basic ideas from the TAPP group as they were presented in the SRI Business Executive Seminars from 1965 – 1971 where he was the principal trainer and consultant. The first phase is called participative planning where the strengths, weaknesses, opportunities and threats inherent in the operation need to be identified. Both articles provide no names or references about the methodology or what this step actually tries to accomplish. This confirms the inverse relation between decoupling and the length of the incubation period (Røvik 2011).

In 1972 Humphrey started his own consultancy firm called Business Planning & Development in London and developed the Team Action Management method. His firm had a lot of competition from Urwick Orr & Partners who also made their own version of the SWOT analysis. The longer a managerial idea lingers at the discourse level, the more likely it is that eventual decoupling between idea and practice will be discovered and becomes criticized (Røvik 2011). As we have shown in the introduction, SWOT analysis as a strategy tool is critiqued for this exact reason, but mainly from an scientific viewpoint for lacking methodological rigour. While the value of SWOT is not only seen in its outputs, but also in the very process of carrying it out (Pickton & Wright 1998).

Due to copyrights and limited distribution, the original application of SWOT analysis starts to fade and never reaches the academic literature, which makes it hard to make claims about *maturation* in practice in the absence of data.

Mutation of the SOFT approach

The fifth salient virus feature is mutation. Changes in the genetic structure may occur due to irregularities in the replication process (Røvik 2011). Two year later after the first SOFT approach publication in LRPS no. 250 (Stewart, Benepe & Dosher 1965), objectives, strengths and weaknesses, opportunities and threats, performance projections, are again brought together in a new evaluation of the prospects for the firm. The actual procedures for startup consist of four phases; phase 1: planning issues, phase 2: Structuring SOFT, phase 3: Gap analysis and phase 4: Provisional plan (Stewart, Humphrey & Smith 1965). In an effort to determine the discrepancy between the desired and the probable future of the firm's present products and markets. In the advanced stage called “entrepreneurial planning”, the firm sets corporate objectives, examines its strengths and weaknesses, probes deeply for external opportunities and threats, and-combining all of these - makes a systematic evaluation of its prospects. Any proposals for change undergo intensive search and analysis, culminating in an action decision, which then enters the flow pattern established in earlier stages of planning. Entrepreneurial Planning represents a major commitment of the firm's resources and top management time and can altogether alter the organization and atmosphere of the company (Ansoff & Stewart 1967).

In Setting Corporate Aims (Stewart 1971), the last LRPS report before retiring in 1971, Stewart revisited his original agenda for research, developed at Lockheed and largely implemented in his time at SRI, which called for a ‘three-fold attack on systematizing the body of knowledge relating to corporate planning’, which up to then was considered strictly an art (Royce 1985; Stewart 1971 and McConnell 1971). An observation which was later repeated (Steiner 1979).

The sequence of the focus was intended to be:

1. The Planning Process (The domain of TAPP from 1962 to 1971)
2. The Creative Thinking Process (initiated by LRPS/TAPP)
3. The Corporate Development Function – inventing new business

This combination of activities, including a related work on stakeholder values, intelligence acquisition and management, decision analysis, and implementation planning, comprised the domain now covered by ‘Strategic Management’, or ‘Organized Entrepreneurship’ as Stewart and his colleagues dubbed it in 1966 (Royce 1985). SOFT approach *mutated* to SWOT analysis.

Cognitive artefact	Year	Author(s)	Outlet
SOFT approach	1965	Stewart, Benepe & Dosher	Report
SWOT analysis	1967	Ansoff & Stewart	Report
SWOT analysis	1972	Stait	Industrial and commercial training
WOTS-Up	1977	Steiner & Miner	Book
T/O/S/W	1980	Ansoff	Strategic Management Journal
TOWS matrix	1982	Weihrich	Long Range Planning

Table 5 Cognitive artefacts - Historical variations on SOFT approach 1965 - 1982

Dormancy

The sixth salient virus feature is dormancy. The key organizational processes in the observed pattern of dormancy, are inactivation and reactivation (Røvik 2011). An interesting phenomenon over time is the need to circumvent the lack of clear instructions for use and outcomes measures by making hybrid SWOT analysis models. This is *reactivation* of a dormant idea with an iteration of the maturation and translation mechanisms (see table 5). SWOT analysis has now crossed over to other domains, the so-called cross species host-host interactions. SWOT crossed over from Business domain to Psychology, Technology and even the Medical domains (see table 6).

Cognitive artefact	Year	Author(s)	Outlet
SWOT-AHP	2000	Kurtilla, Personen, et al.	Forest Policy and Economics
A'WOT	2001	Kangas, Personen, et al.	Conference Proceedings
Delphi-SWOT	2012	Tavana, Pirdashti, et al.	Energy Policy
SWOT-FANP	2013	Lee	Technological and Economic Development of Economy
SORF analysis	2013	Shadfar	Journal of Marketing and Management
ANP-SWOT	2014	Shahabi	Journal of Facilities Management
SWUF analysis	2016	Siciliano	Journal of the Academy of Business Education
I-SWOT	2017	Sachweh et al.	Gefasschirurgie

Table 6 Cognitive artefacts - Hybrids on SWOT analysis 2000-2017

Discussion

The SOFT approach was designed to solve the so-called planning paradox. Participative planning became key by involving stakeholders (Freeman & Reed 1983; Stewart, Allen & Cavender 1963) and working within the so-called chain of reasoning (Stewart 1965) to design the planning for both short- and long term operations. Our endeavour illustrates that the virus theory perspective provides much richer details on the evolution of a cognitive artefact from both an organizational and institutional level. The role of practitioners deserves much more attention when discussing the evolution and diffusion of management ideas. The management fashion theory clearly has an academic bias which inhibits our understanding of the institutionalization of management ideas.

The TAPP group where reflective and practiced evidence-based management (Rousseau 2012) *avant la letter*. Robert F. Stewart was considered the business guru who set the research agenda. In the seminars, practical planning issues were addressed and solutions where provided. Practitioners were in the lead in the infection and diffusion of the SWOT analysis in practice.

From our research we know that management consultants from Arthur D. Little, McKinsey & Company and Ernst & Ernst (SRI 1970) have participated in the SRI Executive Business Seminars. We also know that two consultants from Urwick Orr & Partners Ltd. attended (Royce 2001). Immunity reactions were triggered due to tuition fees, which inhibited the uptake in an MBA courses (for instance at Harvard Business School and UCLA). The mutation of management ideas proves to be crucial is the dissemination of management ideas, where remnants of the original management ideas stay alive or live on in another form. After a dormancy of almost 18 years, the SWOT mnemonic is reactivated and hybrid forms appear in scholarly journals. We see that authors from other disciplines try to circumvent the lack of outcome measure (in the absence of the theoretical underpinnings). We suggest an assessment the dominant strategy text-books in the field to understand the consequences (and possibly update them).

Conclusions

SOFT approach was the original concept by Robert F. Stewart and TAPP group, incorporated in Formal Planning - The Staff Planner's Role at Start Up Planning – LRPS report no. 250. (Stewart, Benepe & Dosher 1965) and in SRI seminar work. Two years later is was renamed SWOT analysis in The Evolution of Corporate Planning – LRPS report no. 329 (Ansoff & Stewart 1967).

The LRPS reports are copyright protected and where in limited circulation (only for subscribers to the Long Range Planning Service).This could explain why the first use of SWOT analysis as mnemonic is found in the article Management Training and the Smaller Company (Stait 1972) from a senior partner at Urwick Orr & Partners Ltd. in London. Some Urwick Orr & Partners clients followed courses on Improving Management Performance (IMP). We have found a follow up report called Improving management performance: a guide to “SWOT” analysis from the Chemical & Allied Product ITB training board (Mottershaw 1974) in Worldcat. In the book Urwick, Orr on Management (Latham & Sanders 1980) they present their own version of “SWOT” analysis, again with no references. The process steps which are described in LRPS no. 250 (Stewart, Benepe & Dosher 1965) are missing.

Our study demonstrated the theoretical underpinnings of the SWOT analysis disappeared in the literature. The SOFT approach has gradually mutated in practice (table 8). Variations ranging as wordplay to conceptual modification.

The oldest publication on a 2 x 2 SWOT like matrix as cognitive artefact is found in Strategic Issues (Ansoff 1980) in the Strategic Management Journal.

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Appendix

1. (April 2015). Requested the report A Guide to "SWOT analysis – Strength, Weaknesses, Opportunities, Threats (Mottershaw 1974) which the authors found in Worldcat from the National Library of Ireland.
2. (June 2016). Email contact with dr. James Matherson, who founded SmartOrg.com and previously headed the Decision Analysis program at SRI for 15 years. He sent us PowerPoint presentations about his work at SmartOrg.com and his work as consulting professor at Stanford University.
3. (August 2016). Meeting with professor William D. Guth from the New York University Stern School of Business at the annual meeting of the Academy of Management in Anaheim, California.
4. (August 2016). The authors visited Strategic Business Insights (SBI) in Menlo Park and where able to peruse an old LRPS box from the SBI archive with seminar material and LRPS reports from the TAPP group. The authors received the internal report A History of Strategic Management and Planning at SRI (Royce 1985). We also visited SmartOrg.com and met with Don Creswell, dr. Peter McNamara and dr. Carl Spetzler. No new information emerged.
5. (February 2017). Reached out via social media (Facebook) to Doreen Mortimer and Derek Hindley, who both used to work as consultants for Urwick Orr & Partners in London between 1970 and 1990. Doreen Mortimer had used to have a UOP Pensions group dating back to 2009, but in 2010 the group disbanded. The Urwick Management Center was closed in 1990 due to a merger with Price Waterhouse and the archives have been destroyed. She referred us to the book Urwick, Orr & Partners (Latham & Sanders 1980), which we bought on Amazon.com.
6. (March 2017). Requested information to the Library of Congress in March 2017 to find out if hardcopies of the Long Range Planning reports, which have been filed for copyrights in the online Catalogues of Copyrights between 1965 – 1975 are available on microfilm or in hardcopy. After several request the authors were informed that there was the requested information was misplaced and currently is not available.
7. (May 2017). Confirmation of three transcripts from the Google Hangout/Skype interviews by professor William D. Guth. Dates: November 11th 2016; November 12th 2016 and December 7th. Two informal interviews via Google Hangout on Air recordings were made on May 11th 2017 and May 31 th 2017.
8. (May 2017). Requested and received hardcopies from the Kenneth Andrews Papers in the Baker Library, Series IV. Harvard Business School and Harvard University Conferences, Symposia, etc. 1946-1993.
    a. Box 13, Folder 21 Stanford University
    b. Box 14, Folder 13 Urwick, Orr & Partners Ltd.
    c. Box 72, Folder 7 HBS: Business Symposium: April 7-11: Correspondence, 1963-1969
9. Catalogued and structured the personal archive of one of the authors with proprietary SRI material with reports, seminar material and personal correspondence in an MS Access 2013 database. An overview of the whole dataset will be published in figshare.